

[531] A “SMITH’S PRIZE” PAPER; SOLUTIONS (continued)

p. 442 [531]

where $\frac{d\xi}{dt}$, $\frac{d\eta}{dt}$ are the velocities of the ball and cannon respectively at the instant of emergence; $\int_0^t \phi(x+S)(v+dS)$, is a constant factor in a quantity independent of the weights of the ball and cannon), and putting it = $\frac{1}{2}V$, the equation may be written

$$(v+V)^2 = \left(\frac{1}{m} + \frac{1}{M}\right) C,$$

where v , V are the velocities at the moment of emergence.

But from the equation

$$m \frac{d^2x}{dt^2} = M \frac{d^2X}{dt^2},$$

(the initial values of the velocities being each = 0), we have

$$mv = MV;$$

and hence from the foregoing equation we obtain

$$v = \frac{M}{M+m}(V+v) = \frac{M}{M+m} \frac{\sqrt{(M+m)C}}{\sqrt{Mm}} = \frac{\sqrt{M}}{\sqrt{m}\sqrt{M+m}} \sqrt{C} = \frac{\sqrt{C}}{\sqrt{m}\sqrt{1+\frac{m}{M}}},$$

or say

$$v\sqrt{m} = \frac{\sqrt{C}}{\sqrt{1+\frac{m}{M}}}.$$

And similarly for the other cannon, if m_1 , M_1 be the masses of the ball and cannon, v_1 the velocity of the ball, we have

$$v_1\sqrt{m_1} = \frac{\sqrt{C}}{\sqrt{1+\frac{m_1}{M_1}}},$$

whence

$$v\sqrt{m} : v_1\sqrt{m_1} = \sqrt{1+\frac{m_1}{M_1}} : \sqrt{1+\frac{m}{M}}.$$

If $m_1 = m$, $M_1 = \alpha$, then v , v_1 may be taken to be the velocities of the same ball fired from the cannon, mass M , when it is free to recoil and when it is absolutely fixed; and we have

$$v : v_1 = 1 : \sqrt{1+\frac{m}{M}},$$

viz. the velocity in the former case is equal to that in the latter case divided by

$$\sqrt{1+\frac{m}{M}}.$$

p. 443 [531]

4. If $X : Y : Z = \eta\zeta' - \eta'\zeta : \zeta\xi' - \zeta'\xi : \xi\eta' - \xi'\eta$, where $\xi^2 + \eta^2 + \zeta^2 = 0$, $\xi'^2 + \eta'^2 + \zeta'^2 = 0$, show that $\xi\xi'$, $\eta\eta'$, $\zeta\zeta'$, $\eta\zeta' + \eta'\zeta$, $\zeta\xi' + \zeta'\xi$, $\xi\eta' + \xi'\eta$ are proportional to quadric functions of X , Y , Z .

We have

$$\begin{aligned} Y^2 + Z^2 &= (\zeta\xi' - \zeta'\xi)^2 + (\xi\eta' - \xi'\eta)^2 \\ &= \xi'^2(\eta^2 + \zeta^2) + \xi^2(\eta'^2 + \zeta'^2) - 2\xi\xi'(\eta\eta' + \zeta\zeta') \end{aligned}$$

which, in virtue of the equations

$$\xi^2 + \eta^2 + \zeta^2 = 0, \quad \xi'^2 + \eta'^2 + \zeta'^2 = 0,$$

is

$$\begin{aligned} &= -2\xi^2\xi'^2 - 2\xi\xi'(\eta\eta' + \zeta\zeta') \\ &= -2\xi\xi'(\xi\xi' + \eta\eta' + \zeta\zeta'). \end{aligned}$$

And again

$$\begin{aligned} YZ &= (\zeta\xi' - \zeta'\xi)(\xi\eta' - \xi'\eta) \\ &= \xi\xi'(\eta\zeta' + \eta'\zeta) - \xi'^2\eta\zeta - \xi^2\eta'\zeta', \end{aligned}$$

which, in virtue of the same equations, is

$$\begin{aligned} &= \xi\xi'(\eta\zeta' + \eta'\zeta) + \eta\zeta(\eta'^2 + \zeta'^2) + \eta'\zeta'(\eta^2 + \zeta^2) \\ &= (\eta\zeta' + \eta'\zeta)(\xi\xi' + \eta\eta' + \zeta\zeta'), \end{aligned}$$

and, forming the analogous equations by symmetry, the factor $(\xi\xi' + \eta\eta' + \zeta\zeta')$ divides out, and we have

$$\begin{aligned} Y^2 + Z^2 : Z^2 + X^2 : X^2 + Y^2 : -2YZ : -2ZX : -2XY \\ = \xi\xi' : \eta\eta' : \zeta\zeta' : \eta\zeta' + \eta'\zeta : \zeta\xi' + \zeta'\xi : \xi\eta' + \xi'\eta, \end{aligned}$$

which is the required theorem.

5. *Two tangents of a conic are harmonically related to a second conic; find the locus of the intersection of the two tangents.*

In plane geometry the angle in a semicircle is, in spherical geometry it is not, a right angle: show how these conclusions follow from the solution of the above problem.

Let the equation of the first conic be $x^2 + y^2 + z^2 = 0$; its equation in line coordinates is therefore $u^2 + v^2 + w^2 = 0$; or what is the same thing, the line $ux + vy + wz = 0$ will be a tangent of the conic if only $u^2 + v^2 + w^2 = 0$. Hence the equations of the two tangents being

$$\xi x + \eta y + \zeta z = 0,$$

$$\xi' x + \eta' y + \zeta' z = 0,$$

we have $\xi^2 + \eta^2 + \zeta^2 = 0$, $\xi'^2 + \eta'^2 + \zeta'^2 = 0$, and X , Y , Z the coordinates of the point of intersection of these tangents, we have

$$X : Y : Z = \eta\zeta' - \eta'\zeta : \zeta\xi' - \zeta'\xi : \xi\eta' - \xi'\eta,$$

p. 444 [531]

and thence, by the last question,

$$\begin{aligned} & \xi\xi' : \eta\eta' : \zeta\zeta' : \eta\zeta' + \eta'\zeta : \zeta\xi' + \zeta'\xi : \xi\eta' + \xi'\eta \\ & = Y^2 + Z^2 : Z^2 + X^2 : X^2 + Y^2 : -2YZ : -2ZX : -2XY. \end{aligned}$$

Suppose that the equation of the second conic in line coordinates is

$$(a, b, c, f, g, h)(u, v, w)^2 = 0,$$

the condition in order that the two lines $\xi x + \eta y + \zeta z = 0$, $\xi' x + \eta' y + \zeta' z = 0$, may be harmonically related to this conic is

$$(a, b, c, f, g, h)(\xi, \eta, \zeta)(\xi', \eta', \zeta') = 0;$$

or, what is the same thing, it is

$$a\xi\xi' + b\eta\eta' + c\zeta\zeta' + f(\eta\zeta' + \eta'\zeta) + g(\zeta\xi' + \zeta'\xi) + h(\xi\eta' + \xi'\eta) = 0,$$

or, substituting the proportional quadric functions found above,

$$a(Y^2 + Z^2) + b(Z^2 + X^2) + c(X^2 + Y^2) - 2fYZ - 2gZX - 2hXY = 0,$$

or, what is the same thing,

$$(a + b + c)(X^2 + Y^2 + Z^2) - (a, b, c, f, g, h)(X, Y, Z)^2 = 0,$$

as the locus of the point of intersection of the two conics; the required locus is therefore a conic.

It is to be observed that the locus is that of a point which is such that the pairs of tangents from it to the two given conics respectively form a harmonic pencil; viz. if the equations of the given conics (in line coordinates) are

$$u^2 + v^2 + w^2 = 0,$$

$$(a, b, c, f, g, h)(u, v, w)^2 = 0;$$

then the equation of the required locus, or say the equation of the harmonic conic, is (in point coordinates)

$$(a + b + c)(x^2 + y^2 + z^2) - (a, b, c, f, g, h)(x, y, z)^2 = 0.$$

In particular, if the second conic be a point-pair or, say, if its equation be

$$(\alpha u + \beta v + \gamma w)(\alpha' u + \beta' v + \gamma' w) = 0,$$

then the equation of the harmonic conic is

$$(\alpha\alpha' + \beta\beta' + \gamma\gamma')(x^2 + y^2 + z^2) = (\alpha x + \beta y + \gamma z)(\alpha' x + \beta' y + \gamma' z),$$

which is satisfied by writing $(x, y, z) = (\alpha, \beta, \gamma)$, or $= (\alpha', \beta', \gamma')$, viz. the harmonic conic passes through the two points of the point-pair. And so, if the first conic be a point-pair, the harmonic conic passes through the four points of the two point-pairs.

p. 445 [531]

The equation of the first conic in point coordinates is $x^2 + y^2 + z^2 = 0$; hence, supposing as above, that the second conic is a point-pair, the intersections of the first conic with the harmonic conic are given by

$$x^2 + y^2 + z^2 = 0, \quad (\alpha x + \beta y + \gamma z)(\alpha' x + \beta' y + \gamma' z) = 0;$$

whence the harmonic conic has double contact with the first conic, only if each of the lines $\alpha x + \beta y + \gamma z$, $\alpha' x + \beta' y + \gamma' z = 0$, touches the first conic; or, what is the same thing, if each of the points (α, β, γ) , $(\alpha', \beta', \gamma')$ lies on the first conic.

In plane geometry, we have (in the plane) a point-pair, the two circular points at infinity; any conic through these points is a circle: the two lines harmonic in regard hereto are at right angles. Hence, by what precedes, taking one of the given conics to be the circular points at infinity, and the other conic to be any two points P, Q ; the locus of the intersection of lines through P, Q , cutting each other at right angles, is a circle; this is evidently the circle standing on PQ as diameter, or the angle in a semicircle is a right angle.

In spherical geometry we have (on the sphere) an imaginary conic $x^2 + y^2 + z^2 = 0$, called the *absolute*; any conic having double contact herewith is a circle (small circle of the sphere); two lines (arcs of great circles), harmonic in regard hereto, are at right angles. Hence, taking one of the conics to be the conic $x^2 + y^2 + z^2 = 0$ and the other to be the two points P, Q ; the locus of the intersections of the lines (arcs of great circles) through P, Q , which cut at right angles, is a spherical conic; but it is not a circle unless the points P, Q , are each of them on the conic $x^2 + y^2 + z^2 = 0$, viz. it is not a circle for any two real points whatever; that is, in spherical geometry, the angle in a semicircle is not a right angle.

6. *A mass M attached to the end A of a chain AC is placed (with the chain) on a horizontal plane, in such wise that a portion AB of the chain forms a straight line, the remaining portion BC being heaped up at B : the mass M is then set in motion in the direction B to A with a given velocity, and so moves in a straight line, dragging the chain: determine the motion; and explain the peculiarity of the dynamical problem.*

The mass attached to the end of the chain is taken to be M , and the mass of a unit of length of the chain to be $= m$; suppose also that at the commencement of the motion the distance CA is $= a$, and that at the end of the time t this distance is $= a + x$, so that the length of chain then in motion is $= a + x$, ($a < l - a$, if l be the whole length of the chain). Suppose also that the velocity at the time t is $= v$; then we have a mass $= M + m(a + x)$ moving with a velocity v ; and, in the element of time dt , this sets in motion with the velocity v a length of chain $dx = vdt$, or mass of chain $= mvdt$; if then the impulse backwards on the mass $M + m(a + x)$, and forwards on the element $mvdt$ be $= R$, we have

$$\{M + m(a + x)\} dv = -R,$$

$$mvdt \cdot v = R,$$

p. 446 [531]

that is,

$$[M + m(a + x)] dv + mv^2 dt = 0;$$

or, observing that $v dt = dx$, this may be written

$$(M + ma) dv + m d.xv = 0,$$

that is,

$$\begin{aligned} [M + m(a + x)]v &= \text{constant}, \\ &= (M + ma)V, \end{aligned}$$

if V be the velocity at the commencement of the motion. The equation gives, in terms of the space x , the velocity v at any time t before the whole chain is set in motion; writing it in the form

$$\{M + m(a + x)\} \frac{dx}{dt} = (M + ma)V,$$

we have

$$(M + ma)x + \frac{1}{2}mx^2 = (M + ma)Vt,$$

and putting herein $x = l - a$, the equation gives the value of t at the instant when the whole chain is set in motion; after this epoch, the mass and chain will (it is clear) move on with a uniform velocity

$$= \frac{(M + ma)V}{M + ml},$$

The foregoing equation

$$[M + m(a + x)]v = (M + ma)V$$

might have been obtained at once by the consideration that the momentum is constant throughout the motion; but the method employed puts more clearly in evidence the peculiarity of the dynamical problem; viz. it is, so to speak, a problem of continuous impulse: in each element of time dt an infinitesimal element of mass has its velocity abruptly altered (in the present problem from 0 to v), but, for the very reason that it is an infinitesimal element of mass which undergoes this abrupt change of velocity, the effect is a continuous, not an abrupt, change of velocity of the whole finite mass which is then in motion.

7. Show how an ellipse may be constructed as the envelope of a variable circle having its centre upon either of the axes; and examine the geometrical peculiarities which occur according as the major or the minor axis is made use of.

Considering the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1,$$

the equation

$$(x - \alpha)^2 + y^2 = \gamma^2,$$

where α and γ are in the first instance arbitrary parameters, represents a circle having its centre on the major axis: in order that this may touch the ellipse, a relation must be established between α and γ . When this is done we obtain a variable circle

p. 447 [531]

(the equation of which contains a single arbitrary parameter), and this circle has the ellipse for its envelope. (On account of the symmetry in regard to the axis of x , the circle will, it is clear, touch the ellipse in two points, situate symmetrically in regard to this axis.) Now to make the circle touch the ellipse, eliminating y , we have

$$(x - \alpha)^2 + b^2 \left(1 - \frac{x^2}{a^2}\right) - \gamma^2 = 0,$$

that is,

$$x^2 \left(1 - \frac{b^2}{a^2}\right) - 2\alpha x + \alpha^2 + b^2 - \gamma^2 = 0,$$

which equation, considered as an equation in x , must have equal roots; that is, we must have

$$\left(1 - \frac{b^2}{a^2}\right) (\alpha^2 + b^2 - \gamma^2) - \alpha^2 = 0;$$

or, what is the same thing,

$$-b^2\alpha^2 + (a^2 - b^2)(b^2 - \gamma^2) = 0;$$

consequently

$$\gamma^2 = b^2 \left(1 - \frac{\alpha^2}{a^2 - b^2}\right);$$

or the required equation of the circle is

$$(x - \alpha)^2 + y^2 = b^2 \left(1 - \frac{\alpha^2}{a^2 - b^2}\right);$$

or, as this may also be written

$$(x - \alpha)^2 + y^2 = (1 - e^2) \left(\frac{a^2}{e^2} - \alpha^2\right);$$

It is to be observed that at the points of contact of the ellipse and circle, we have

$$x = \frac{a^2}{a^2 - b^2} \alpha = \frac{\alpha}{e^2}.$$

By simply interchanging a and b , it appears that when the centre is on the minor axis, the equation of the variable circle is

$$x^2 + (y - \beta)^2 = b^2 \left(1 - \frac{\beta^2}{b^2 - a^2}\right);$$

and that at the points of contact of the ellipse and circle we have

$$y = \frac{b^2}{b^2 - a^2} \beta.$$

In the case where the point is on the major axis, then attending to positive values of α , the circle remains real so long as α is $< ae$, but if α is $= ae$, that is, if the centre be at the focus, the radius of the circle is $= 0$. But from the formula $x = \alpha/e^2$, we have $x = a$ for $\alpha = ae^2$ and, when α becomes greater than this, the points of contact are imaginary. That is, as α passes from 0 to ae^2 , the circle is real, and

p. 448 [531]

has real contact with the ellipse; for $\alpha = ae^2$ the circle becomes the circle of maximum curvature, touching the ellipse at the extremity of the major axis; as α increases from ae^2 to ae , the circle is still real but its contact with the ellipse is imaginary; for $\alpha = ae$, the circle reduces itself to the focus considered as an evanescent circle; and for greater values of α , the circle is imaginary.

When the centre is on the minor axis it appears in like manner that the circle is always real; but, attending to negative values of β , the circle has real contact with the ellipse only so long as $\beta > b - \frac{a^2}{b}$; for this value of β , the circle touches the ellipse at the extremity of the minor axis, being in fact the circle of minimum curvature; and for greater negative values of β , the contact is imaginary.

8. *Show that the number of ways in which n things can be arranged so that no one of them occupies its original position is of the form $(n-1)A_n$; and that we have $A_1 = 0$, $A_2 = 1$, $A_n = (n-2)A_{n-1} + (n-3)A_{n-2}$: show also that*

$$A_n = (n-1)A_{n-1} - \frac{1}{n-1}\{A_{n-1} + (-)^{n-1}1\}.$$

Supposing the n things arranged as required, we may by a single transposition of two things bring a given thing, say n , into its original place (the last place): and conversely every arrangement of the required form can be obtained from an arrangement in which n occupies the last place, by a transposition of n with another of the things. Now in the arrangement which thus gives rise to an arrangement of the required form, either all the things $1, 2, 3, \dots, (n-1)$ are out of their original places; and we can then transpose n with any one of the things $1, 2, 3, \dots, n-1$: or else only one thing is in its original place; say 1 is in its original place, and we can then transpose 1 and n ; or 2 is in its original place, and we can transpose 2 and n ; $\dots$ or $n-1$ is in its original place, and we can transpose $n-1$ and n . And these are the only ways in which an arrangement of the required form is obtained. Hence if for n things we denote by U_n the number of arrangements in which no one thing occupies its original place, we have, by what precedes,

$$U_n = (n-1)(U_{n-1} + U_{n-2}),$$

and it thus appears that U_n is of the form $(n-1)A_n$; and writing accordingly $U_n = (n-1)A_n$, and therefore

$$U_{n-1} = (n-2)A_{n-1}, \quad U_{n-2} = (n-3)A_{n-2},$$

we have

$$A_n = (n-2)A_{n-1} + (n-3)A_{n-2};$$

which is the required equation for A_n ; we have, it is clear, $A_1 = 0$, $A_2 = 1$, and the successive values $A_3 = 1$, $A_4 = 3$, &c. can then be calculated. But the equation of differences of the second order may be integrated into one of the first order, viz. writing the equation in the form

$$\{(n-1)A_n - n(n-2)A_{n-1}\} + \{(n-2)A_{n-1} - (n-1)(n-3)A_{n-2}\} = 0,$$

p. 449 [531]

the integral is

$$(n-1)A_n - n(n-2)A_{n-1} = (-)^{n-1}C,$$

and writing $n = 2$, we have $C = -1$, wherefore

$$(n-1)A_n = n(n-2)A_{n-1} - (-)^{n-1}1,$$

or, what is the same thing,

$$A_n = (n-1)A_{n-1} - \frac{1}{n-1}\{A_{n-1} + (-)^{n-1}1\},$$

which is the other equation for A_n .

The foregoing very elegant proof of the equation

$$U_n = (n-1)(U_{n-1} + U_{n-2}) = (n-1)A_n$$

was unknown to me; but was given in the Examination; my own proof of the equation $U_n = (n-1)A_n$ was derived from the well-known formula

$$U_n = 1 \cdot 2 \cdots n \left\{ 1 - \frac{1}{1} + \frac{1}{1 \cdot 2} - \cdots + (-)^n \frac{1}{1 \cdot 2 \cdot 3 \cdots n} \right\}.$$

The theorem itself, and the two equations of differences for A_n are due to Euler, see his memoir, "Sur une espèce particulière de Carrés Magiques," *Comm. Arith. Coll.*, t. I., p. 359.

9. If in any covariant of a binary form $(a, b, c, \dots, k)(x, y)^n$ the coefficients $a, b, \dots, k$ are replaced by $ax + by, bx + cy, \dots, kx + ly$ respectively, show that the result is a covariant of the next superior form $(a, b, c, \dots, l)(x, y)^{n+1}$: and determine the covariant obtained by thus operating on the discriminant of the cubic form $(a, b, c, d)(x, y)^3$.

A covariant of the binary form $U = (a, b, \dots, k)(x, y)^n$ is either given by a single symbolical expression of the form $\overline{12}^a \overline{13}^\beta \dots \overline{23}^\gamma \dots U_1 U_2 U_3 \dots$, or it is the sum of a number of such expressions, each multiplied by a constant numerical factor. In the latter case each of the expressions in question is a covariant of U . Any such expression is at once seen to be a function of U and of its differential coefficients (i.e. it may contain the differential coefficients of each or any of the orders $0, 1, 2, \dots, n$), homogeneous as regards the differential coefficients of the same order. But writing $U' = (a, b, c, \dots, l)(x, y)^{n+1}$, the differential coefficients of any order of U are by the change of $a, b, \dots, k$ into $ax + by, bx + cy, \dots, kx + ly$, converted into the differential coefficients of the same order of U' , each multiplied into the same merely numerical coefficient; the result (disregarding numerical factors) is thus the same derivative $\overline{12}^a \overline{13}^\beta \dots \overline{23}^\gamma \dots U'_1 U'_2 U'_3 \dots$ of U' ; viz. it is a covariant of U' ; and making the like change in any sum of such expressions (each into a numerical factor), the result is a like sum of expressions referring to U' ; that is, making the change in any covariant of U , the result is a covariant of U' .

p. 450 [531]

Or the same thing may be otherwise proved thus; if to fix the ideas we write $U = (a, b, c, d)(x, y)^3$, any covariant Θ of U is reduced to zero by each of the operations

$$a\delta_b + 2b\delta_c + 3c\delta_d - y\delta_x, \quad 3b\delta_a + 2c\delta_b + d\delta_c - x\delta_y;$$

and, conversely, any function Θ which is thus reduced to zero is a covariant of U . Attending to the first operator, we have for any covariant Θ of U ,

$$(a\delta_b + 2b\delta_c + 3c\delta_d - y\delta_x)\Theta = 0.$$

Write $ax + by = a'$, $bx + cy = b'$, $cx + dy = c'$, $dx + ey = d'$, and let Θ' be the function obtained from Θ by changing a, b, c, d into a', b', c', d' . We ought to have

$$(a\delta_b + 2b\delta_c + 3c\delta_d + 4d\delta_e - y\delta_x)\Theta' = 0,$$

or, considering Θ' as a function of a', b', c', d', x, y , this is

$$\{a(y\delta_{a'} + x\delta_{b'}) + 2b(y\delta_{b'} + x\delta_{c'}) + 3c(y\delta_{c'} + x\delta_{d'}) + 4dy\delta_{d'} - y(a\delta_{a'} + b\delta_{b'} + c\delta_{c'} + d\delta_{d'}) - y\delta_x\}\Theta' = 0,$$

or, what is the same thing,

$$\{(ax + by)\delta_{b'} + 2(bx + cy)\delta_{c'} + 3(cx + dy)\delta_{d'} - y\delta_x\}\Theta' = 0,$$

that is,

$$\{a'\delta_{b'} + 2b'\delta_{c'} + 3c'\delta_{d'} - y\delta_x\}\Theta' = 0,$$

an equation which, Θ' being the same function of a', b', c', d', x, y that Θ is of a, b, c, d, x, y , is satisfied identically; and thus Θ' is reduced to zero by the operation $a\delta_b + 2b\delta_c + 3c\delta_d + 4d\delta_e - y\delta_x$; and similarly it is reduced to zero by the operation $4b\delta_a + 3c\delta_b + 2d\delta_c + e\delta_d - x\delta_y$; and it is thus a covariant of $(a, b, c, d, e)(x, y)^4$.

The discriminant of $(a, b, c, d)(x, y)^3$ is

$$\Theta = a^2d^2 + 4ac^3 + 4b^3d - 6abcd - 3b^2c^2;$$

making the substitution in question, it is

$$= (ax + by)^2(dx + ey)^2 + \&c.,$$

viz. it is

$$= (a^2d^2 + 4ac^3 + 4b^3d - 6abcd - 3b^2c^2)x^4 + \&c.$$

But there is no such irreducible covariant of the quartic function $(a, b, c, d, e)(x, y)^4$; and observing that we have identically

$$(ae - 4bd + 3c^2)(ac - b^2) - (ace - ad^2 - b^2e - c^3 + 2bcd)a = a^2d^2 + 4ac^3 + 4b^3d - 6abcd - 3b^2c^2,$$

the covariant in question must be

$$= (ae - 4bd + 3c^2)\{(ac - b^2)x^4 + \&c.\} - (ace - ad^2 - b^2e - c^3 + 2bcd)(a, b, c, d, e)(x, y)^4,$$

where $(ac - b^2)x^4 + \&c.$ is the Hessian of the quartic function $(a, b, c, d, e)(x, y)^4$.

10. *From the equation of the curves of curvature of an ellipsoid, or otherwise, determine the curves of curvature of a paraboloid: show also that for the paraboloid $xy = cz$ (the parallel sections $z = \text{Const.}$ being rectangular hyperbolas) the curves of curvature are the intersections of the paraboloid by the system of surfaces*

$$h = \sqrt{x^2 + z^2} \pm \sqrt{y^2 + z^2}.$$

p. 451 [531]

The curves of curvature of the ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1,$$

are given as the intersection of the surface with the confocal surface

$$\frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} + \frac{z^2}{c^2 + \lambda} = 1;$$

transforming to the vertex, or writing $z - c$ in place of z , these become

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2 - 2cz}{c^2} = 0,$$

$$\frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} + \frac{z^2 - 2cz - \lambda}{c^2 + \lambda} = 0,$$

or multiplying by c and writing $\frac{a^2}{c} = l$, $\frac{b^2}{c} = m$, $\frac{\lambda}{c} = \theta$, the equations are

$$\frac{x^2}{l} + \frac{y^2}{m} + \frac{z^2}{c} - 2z = 0,$$

$$\frac{x^2}{l + \theta} + \frac{y^2}{m + \theta} + \frac{z^2 - 2z + \theta}{1 + \frac{\theta}{c}} = 0,$$

or, putting herein $c = \infty$, the equations are

$$\frac{x^2}{l} + \frac{y^2}{m} - 2z = 0,$$

$$\frac{x^2}{l + \theta} + \frac{y^2}{m + \theta} - 2z - \theta = 0,$$

viz. these equations, where θ is a variable parameter, determine the curves of curvature of the paraboloid $\frac{x^2}{l} + \frac{y^2}{m} - 2z = 0$. The second equation may be replaced by

$$\frac{x^2}{l(l + \theta)} + \frac{y^2}{m(m + \theta)} + 1 = 0.$$

Write in the equations $l = -m = c$; they become

$$x^2 - y^2 - 2cz = 0,$$

$$\frac{x^2}{c + \theta} + \frac{y^2}{c - \theta} + c = 0;$$

or, substituting herein $\frac{x + y}{\sqrt{2}}$ for x , and $\frac{x - y}{\sqrt{2}}$ for y , the equations are

$$xy - cz = 0,$$

$$\frac{(x + y)^2}{c + \theta} + \frac{(x - y)^2}{c - \theta} + 2c = 0,$$

p. 452 [531]

the second of which is

$$c(x^2 + y^2) - 2\theta xy + (c^2 - \theta^2) = 0,$$

which is the equation for determining the curves of curvature of the surface $xy - cz = 0$.

Writing in this equation $\theta = \sqrt{h^2 + c^2}$, it becomes

$$c(x^2 + y^2) - 2xy\sqrt{h^2 + c^2} - ch^2 = 0,$$

or, as this may be written,

$$(c^2 + x^2)(c^2 + y^2) - x^2y^2 - 2cxy\sqrt{h^2 + c^2} - c^2h^2 - c^4 = 0,$$

that is,

$$\pm\sqrt{c^2 + x^2}\sqrt{c^2 + y^2} = xy + c\sqrt{h^2 + c^2},$$

and thence

$$\begin{aligned} c^2h^2 &= (c^2 + x^2)(c^2 + y^2) + x^2y^2 - c^4 \mp 2xy\sqrt{c^2 + x^2}\sqrt{c^2 + y^2} \\ &= x^2(c^2 + y^2) + y^2(c^2 + x^2) \mp 2xy\sqrt{c^2 + x^2}\sqrt{c^2 + y^2}, \end{aligned}$$

that is,

$$ch = x\sqrt{c^2 + y^2} \pm y\sqrt{c^2 + x^2};$$

or combining with the equation $xy - cz = 0$ of the paraboloid, this is

$$h = \sqrt{x^2 + z^2} \mp \sqrt{y^2 + z^2},$$

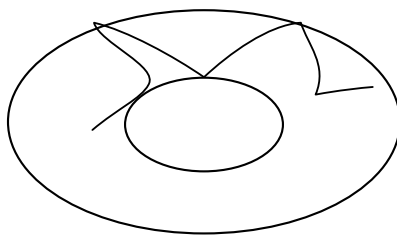
or the curves of curvature are given as the intersections of the paraboloid by the series of surfaces represented by this equation.

11. *Explain for a surface such as the ellipsoid the form of the curve of given constant slope; and in an ellipsoid having one of its principal planes horizontal determine the limits within which the curve is situate.*

At any point of a surface, the direction of the line of greatest slope is evidently at right angles to the level curve, or, what is the same thing, to the trace of the tangent plane on the horizontal plane; and the inclination of the element of the line of greatest slope is equal to that of the tangent plane to the horizontal plane. The line of given constant slope must therefore lie entirely on that portion of the surface for which the inclination of the tangent plane has a value not less than the given constant slope of the curve; viz. on a closed surface such as the ellipsoid, it will (in upon a certain zone of the surface, included between two boundaries, which boundaries are the curves such that at any point thereof the inclination of the tangent plane is equal to the given constant slope; and by what precedes, the direction of the line of given constant slope, at any point where it meets the boundary, is coincident with the direction of greatest slope; viz. it will in general meet the boundary at a finite angle; this implies that the point is a cusp on the curve of given constant slope; and the curve in question will be a curve as shown in the figure, passing continually from one boundary to the other, and where it reaches either boundary, turning back cusp-wise to the other boundary; the curve may in particular cases, after making a circuit, or any number of circuits of the zone, re-enter upon itself, forming a closed

p. 453 [531]

curve; but in general this is not the case, and the curve will go on as above between the two boundaries ad infinitum.



In the case of the ellipsoid, the plane of xy being as usual horizontal, and the equation being

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1,$$

then if λ be the given constant inclination, the boundaries are the series of points at which the inclination of the normal to the axes of z is $= \lambda$; that is, we have

$$\frac{z^2}{c^4} = \left(\frac{x^2}{a^4} + \frac{y^2}{b^4} + \frac{z^2}{c^4} \right) \cos^2 \lambda,$$

or, what is the same thing,

$$\left(\frac{x^2}{a^4} + \frac{y^2}{b^4} \right) \cos^2 \lambda - \frac{z^2}{c^4} \sin^2 \lambda = 0,$$

the boundaries are thus two detached ovals, meeting the principal sections $x = 0$, $y = 0$, in the points given by

$$\frac{z}{y} = \pm \frac{c^2}{b^2} \cot \lambda; \quad \frac{z}{x} = \pm \frac{c^2}{a^2} \cot \lambda,$$

and the general form is thus at once perceived.

12. *To every point of space there corresponds a plane, viz. considering the several points as belonging to a solid body which is infinitesimally displaced in any manner, the plane which corresponds to a point is the plane drawn through the point at right angles to the direction of the motion; determine the plane which corresponds to a given point; and connect the result with any general geometrical theory.*

The displacements, in the directions of the axes, of a point whose coordinates are (x, y, z) are given by the ordinary formulae

$$\delta x = a + qz - ry,$$

$$\delta y = b + rx - pz,$$

$$\delta z = c + py - qx,$$

p. 454 [531]

where a, b, c, p, q, r are constants which determine the particular displacement; hence if X, Y, Z are current coordinates, the plane corresponding to the point (x, y, z) is the plane through this point at right angles to the line

$$\frac{X - x}{\delta x} = \frac{Y - y}{\delta y} = \frac{Z - z}{\delta z},$$

viz. it is the plane

$$(X - x)(a + qz - ry) + (Y - y)(b + rx - pz) + (Z - z)(c + py - qx) = 0,$$

or, substituting for $\delta x, \delta y, \delta z$ their values, reducing and arranging, this is

$$\begin{aligned} &X(a - ry + qz) \\ &+ Y(b - pz + rx) \\ &+ Z(c - qx + py) \\ &+ (-ax - by - cz) = 0, \end{aligned}$$

that is, the equation of the plane corresponding to the point (x, y, z) is an equation linear in regard to the coordinates (x, y, z) of this point; and such that arranging as above in the form of a square, the coefficients which are symmetrically situate in regard to the dexter diagonal are equal and of opposite signs, or say that the coefficients form a skew symmetrical matrix.

More generally, corresponding to the point (x, y, z) we may have the plane

$$AX + BY + CZ + D = 0,$$

where A, B, C, D are any linear functions whatever ($= ax + by + cz + d$, etc.) of the coordinates (x, y, z) ; this is the general relation which is put in the analytical basis of the theory of duality; it is in fact (apart from a variable point situate as a line or a plane, etc.) the general one, to a given point there corresponds a plane not in general passing through the point; the case above considered is distinguished by the circumstance that for any point whatever the corresponding plane does pass through the point.

13. *Write down the Lagrangian equations of motion, explaining the notation, and mode of applying them; and by way of illustration deduce the equations of motion of three particles, connected so as to form an equilateral triangle (of variable magnitude), moving in a plane under the action of any forces.*

The Lagrangian equations of motion are

$$\begin{aligned} \frac{d}{dt} \frac{dT}{d\xi'} - \frac{dT}{d\xi} &= \frac{dU}{d\xi}, \\ \frac{d}{dt} \frac{dT}{d\eta'} - \frac{dT}{d\eta} &= \frac{dU}{d\eta}, \end{aligned}$$

&c.,

p. 455 [531]

$\xi, \eta, \dots$ are here any independent coordinates (in the most general sense of the word) which serve to determine the position of the system at the time t ; T is the *vis viva* function, or half-sum of the mass of each particle of the system into the square of its velocity, expressed in terms of the coordinates $\xi, \eta, \dots$ and of their differential coefficients $\xi' = \frac{d\xi}{dt}$, $\eta' = \frac{d\eta}{dt}$, &c.; T is thus a given function of $\xi, \eta, \dots, \xi', \eta', \dots$, homogeneous of the second order as regards the quantities $\xi', \eta', \dots$. The partial differential coefficients of T in regard to $\xi, \eta, \dots, \xi', \eta', \dots$ are thus given functions of $\xi, \eta, \dots, \xi', \eta', \dots$; U is the force-function or sum

$$\sum m \int (X dx + Y dy + Z dz)$$

(it being assumed that $Xdx + Ydy + Zdz$ is a complete differential, that is, that there exists a force-function U) expressed in terms of the coordinates $\xi, \eta, \dots$ and being thus a given function of these coordinates. The terms $\frac{d}{dt} \frac{dT}{d\xi'}$, &c., contain, it is clear, (and that linearly) the second differential coefficients $\xi'' = \frac{d^2\xi}{dt^2}$, &c.; the equations thus establish between the coordinates $\xi, \eta, \dots$ and their first and second differential coefficients in regard to the time a system of relations, the number of which is equal to that of the coordinates, and which therefore would by integration lead to the expression of the coordinates $\xi, \eta, \dots$ in terms of the time.

It has been tacitly assumed that T, U were functions of $\xi, \eta, \dots, \xi', \eta', \dots$, and of $\xi, \eta, \dots$, not containing the time t ; this is the ordinary case, but there are cases in which T and U or either of them may also contain the time t .

In the proposed case of the three particles, if m_1, m_2, m_3 be their masses, r the side of the equilateral triangle, x, y the coordinates of m_1 , $\theta, \theta + 60^\circ$ (or write for convenience $\theta + a$) the inclinations of m_1m_2, m_1m_3 to the axis of x , then the coordinates of m_2, m_3 will be

$$\begin{aligned} x + r \cos \theta, & \quad x + r \cos(\theta + a), \\ y + r \sin \theta, & \quad y + r \sin(\theta + a). \end{aligned}$$

We have

$$\begin{aligned} T &= \frac{1}{2}m_1(x'^2 + y'^2) \\ &\quad + \frac{1}{2}m_2[(x' + r' \cos \theta - r \sin \theta \theta')^2 + (y' + r' \sin \theta + r \cos \theta \theta')^2] \\ &\quad + \frac{1}{2}m_3[\{x' + r' \cos(\theta + a) - r \sin(\theta + a)\theta'\}^2 + \{y' + r' \sin(\theta + a) + r \cos(\theta + a)\theta'\}^2] \\ &= \frac{1}{2}(m_1 + m_2 + m_3)(x'^2 + y'^2) \\ &\quad + m_2[x'(r' \cos \theta - r \sin \theta \theta') + y'(r' \sin \theta + r \cos \theta \theta')] \\ &\quad + m_3[x'\{r' \cos(\theta + a) - r \sin(\theta + a)\theta'\} + y'\{r' \sin(\theta + a) + r \cos(\theta + a)\theta'\}] \\ &\quad + \frac{1}{2}(m_2 + m_3)(r'^2 + r^2\theta'^2), \end{aligned}$$

p. 456 [531]

a given function of $x, y, r, \theta, x', y', r', \theta'$; also

$$U = m_1 \int (X dx + Y dy) + \&c.$$

will be a given function of x, y, r, θ ; and the equations of motion will be

$$\begin{aligned} \frac{d}{dt} \frac{dT}{dx'} - \frac{dT}{dx} &= \frac{dU}{dx}, \\ \frac{d}{dt} \frac{dT}{dy'} - \frac{dT}{dy} &= \frac{dU}{dy}, \\ \frac{d}{dt} \frac{dT}{dr'} - \frac{dT}{dr} &= \frac{dU}{dr}, \\ \frac{d}{dt} \frac{dT}{d\theta'} - \frac{dT}{d\theta} &= \frac{dU}{d\theta}. \end{aligned}$$

14. *From the integrals $x = a \cos t + a' \sin t$, $y = b \cos t + b' \sin t$, of the dynamical equations $\frac{d^2 x}{dt^2} = -x$, $\frac{d^2 y}{dt^2} = -y$, derive a simultaneous solution of the two partial differential equations*

$$\frac{dS}{dt} + \frac{1}{2} \left\{ \left(\frac{dS}{dx} \right)^2 + \left(\frac{dS}{dy} \right)^2 \right\} = -\frac{1}{2}(x^2 + y^2),$$

$$\frac{dS}{dt} + \frac{1}{2} \left\{ \left(\frac{dS}{da} \right)^2 + \left(\frac{dS}{db} \right)^2 \right\} = -\frac{1}{2}(a^2 + b^2).$$

Writing $U = -\frac{1}{2}(x^2 + y^2)$, we have

$$\begin{aligned} x &= a \cos t + a' \sin t, \\ y &= b \cos t + b' \sin t, \end{aligned}$$

as the integrals of the equations of motion

$$\frac{d^2 x}{dt^2} = \frac{dU}{dx}, \quad \frac{d^2 y}{dt^2} = \frac{dU}{dy};$$

moreover

$$\begin{aligned} x' &= -a \sin t + a' \cos t, \\ y' &= -b \sin t + b' \cos t, \end{aligned}$$

where a, b, a', b' are the initial values of x, y, x', y' respectively $\left(x' = \frac{dx}{dt}, y' = \frac{dy}{dt} \right)$.

Hence the Principal Function

$$\begin{aligned} S &= \int_0^t \frac{1}{2}(T + U) dt \\ &= \frac{1}{2} \int_0^t (x'^2 + y'^2 - x^2 - y^2) dt, \end{aligned}$$

expressed as a function of x, y, a, b, t , will satisfy simultaneously the proposed partial differential equations.

We have

$$x'^2 + y'^2 - x^2 - y^2 = (a'^2 + b'^2 - a^2 - b^2) \cos 2t - 2(aa' + bb') \sin 2t.$$

p. 457 [531]

Hence

$$\begin{aligned} 4S &= \frac{1}{2}(a'^2 + b'^2 - a^2 - b^2) \sin 2t + 2(aa' + bb')(1 - \cos 2t) \\ &= (a'^2 + b'^2) \sin 2t - (a^2 + b^2) \sin 2t - 4(aa' + bb') \sin^2 t. \end{aligned}$$

But

$$\begin{aligned} a' \sin t &= x - a \cos t, \\ b' \sin t &= y - b \cos t. \end{aligned}$$

Hence

$$\begin{aligned} (aa' + bb') \sin t &= ax + by - (a^2 + b^2) \cos t, \\ (a'^2 + b'^2) \sin^2 t &= x^2 + y^2 - 2(ax + by) \cos t + (a^2 + b^2) \cos^2 t, \end{aligned}$$

and substituting these values

$$\begin{aligned} 4S &= \frac{2 \sin t \cos t}{\sin^2 t} \{x^2 + y^2 - 2(ax + by) \cos t + (a^2 + b^2) \cos^2 t\} \\ &\quad - (a^2 + b^2) 2 \sin t \cos t \\ &\quad - 4 \sin t \{ax + by - (a^2 + b^2) \cos t\} \\ &= 2(x^2 + y^2) \frac{\cos t}{\sin t} - \frac{4}{\sin t} (ax + by) + 2(a^2 + b^2) \frac{\cos t}{\sin t}; \end{aligned}$$

or, what is the same thing,

$$S = \frac{1}{2}(x^2 + y^2 + a^2 + b^2) \cot t - 2(ax + by) \operatorname{cosec} t,$$

which value of S satisfies the two partial differential equations.

More generally, the two equations are satisfied by

$$S = c + \text{foregoing value},$$

(c an arbitrary constant) which new value, considered as a solution of the first equation, contains the three arbitrary constants c, a, b , and is thus a *complete* solution; and similarly considered as a solution of the second equation, it contains the three arbitrary constants c, x, y , and is thus a *complete* solution.

I venture to add a few remarks in illustration of what is required in the papers sent up in an Examination.

In the latter part of question (2) (form of the equation for a system of parallel curves) it is worse than useless to say $\frac{df}{dc} \div \left\{ \left(\frac{df}{dx} \right)^2 + \left(\frac{df}{dy} \right)^2 \right\}^{1/2}$ [p. 441] must be constant: a good and sufficient answer would be that it must be constant in virtue of the given equation $f(x, y, c) = 0$. So in question (13) (the Lagrangian equations of motion), it is quite essential to explain [p. 455] that T, U are given functions of $\xi, \eta, \dots, \xi', \eta', \dots$ and of $\xi, \eta, \dots$ respectively; but for this the equations might be partial differential equations for the determination of T, U , or nobody knows what: it is natural and proper to explain further that T is homogeneous of the second order in regard to the derived functions $\xi', \eta', \dots$. In question (14) the answer [p. 456] that $S = \frac{1}{2} \int_0^t (x'^2 + y'^2 - x^2 - y^2) dt$ expressed as a function of x, y, a, b, t will satisfy simultaneously the proposed equations—would be, not of course a complete answer, but a good and creditable one; without the words “expressed as a function of x, y, a, b, t ” it would be altogether worthless. A clear and precise indication of a process of solution is very much better than a detailed solution incorrectly worked out.

[532]

Note on the Integration of Certain Differential Equations by Series.

[From the Oxford, Cambridge, and Dublin Messenger of Mathematics, vol. v. (1869), pp. 77-82.]

THERE is a speciality in the integration of certain differential equations by series, which (though evidently quite familiar to those who have written on the subject—Ellis, Boole, Hargreave) has not, it appears to me, been exhibited in the clearest form. To fix the ideas, consider a linear differential equation of the second order integrable in the form $y = Ax^\lambda + Bx^{\lambda+1} + \dots$; λ is determined by a quadratic equation, and for each value of λ the coefficients $B, C, \dots$ are given multiples of A ; we have thus the general solution

$$y = A \left(x^\alpha + \frac{B}{A} x^{\alpha+1} \dots \right) + K \left(x^\beta + \frac{L}{K} x^{\beta+1} + \&c. \dots \right).$$

The speciality referred to is, when the two roots differ by an integer number; suppose α is the smaller root, and $\beta = \alpha + k$ (k a positive integer) the larger. Then, inasmuch as the series

$$y = Ax^\alpha + Bx^{\alpha+1} + \&c.$$

is identical in form with the general solution as above written down, it is clear that, starting with the root $\lambda = \alpha$, the coefficients beginning with that of $x^\beta, = x^{\alpha+k}$, ought not to be any longer determinate multiples of A , but should contain a new arbitrary constant K , and thus that the series derived from the root $\lambda = \alpha$ should be the general solution containing two arbitrary constants. The most simple case is when the substitution of the series in the differential equation leads to a relation between two consecutive coefficients of the series. Here the values $\frac{B}{A}, \frac{C}{A}, \&c.$ are fractions

p. 459 [532]

the numerators and denominators of which are factorial functions of α such that, for some coefficient $\frac{F}{A}$ preceding $\frac{K}{A}$ (if K is the coefficient of $x^{\alpha+k}$), and for all the succeeding coefficients $\frac{G}{A}$, &c. there is in the numerators one and the same evanescent factor; this being so, it is allowable to write $F = 0$, $G = 0$, &c. giving for the differential equation the finite solution

$$y = A \left(x^\alpha + \frac{B}{A} x^{\alpha+1} \dots + \frac{E}{A} x^{\alpha+e} \right);$$

but, if notwithstanding the evanescent factor, we carry on the series, then in the coefficient of $x^{\alpha+k}$ there occurs in the denominator the same evanescent factor, so that the coefficient of this term presents itself in the form $A \frac{P}{Q} \cdot \frac{0}{0}$, = an arbitrary constant K (since the $\frac{0}{0}$ is essentially indeterminate), and the solution is thus obtained in the form

$$y = A \left(x^\alpha + \frac{B}{A} x^{\alpha+1} \dots + \frac{E}{A} x^{\alpha+e} \right) + K \left(x^{\alpha+k} + \frac{L}{K} x^{\alpha+k+1} + \&c. \right),$$

viz. there is one particular solution which is finite.

Take for example the equation

$$\frac{d^2 y}{dx^2} + q \frac{dy}{dx} - \frac{2}{x^2} y = 0, \quad (\text{I})$$

mentioned Cambridge Math. Journal, t. II. p. 176 (1840). If the integral is assumed to be

$$y = Ax^\alpha + Bx^{\alpha+1} + Cx^{\alpha+2} + \&c.,$$

then we find

$$\begin{aligned} (\alpha + 1)(\alpha - 2)A &= 0, \\ (\alpha - 1)(\alpha + 2)B + q\alpha A &= 0, \\ \alpha(\alpha + 3)C + q(\alpha + 1)B &= 0, \\ (\alpha + 1)(\alpha + 4)D + q(\alpha + 2)C &= 0, \\ &\&c. \end{aligned}$$

Hence $\alpha = -1$, or else $\alpha = 2$;

$$\begin{aligned} B &= \frac{-q\alpha}{(\alpha - 1)(\alpha + 2)} A, \\ C &= \frac{q^2 \alpha(\alpha + 1)}{(\alpha - 1)\alpha(\alpha + 2)(\alpha + 3)} A, \\ D &= \frac{-q^3 \alpha(\alpha + 1)(\alpha + 2)}{(\alpha - 1)\alpha(\alpha + 1)(\alpha + 2)(\alpha + 3)(\alpha + 4)} A, \\ &\&c. \end{aligned}$$

p. 460 [532]

Here taking $\alpha = -1$, we are at liberty to make C and all the following coefficients $= 0$; in fact, if we commence by assuming

$$y = Ax^\alpha + Bx^{\alpha+1},$$

the equations

$$\begin{aligned}(\alpha + 1)(\alpha - 2)A &= 0, \\(\alpha - 1)(\alpha + 2)B + q\alpha A &= 0, \\q(\alpha + 1)B &= 0, \&c.\end{aligned}$$

are all satisfied if only $\alpha = -1$, $B = \frac{-q\alpha A}{(\alpha - 1)(\alpha + 2)}$; and we have thus the finite solution $y = A\left(\frac{1}{x} - \frac{q}{2}\right)$; but if we continue the series, retaining D to represent the indeterminate quantity $\frac{-q^3\alpha(\alpha + 1)(\alpha + 2)}{(\alpha - 1)\alpha(\alpha + 1)(\alpha + 2)(\alpha + 3)(\alpha + 4)}A$, we have the solution

$$y = A\left(\frac{1}{x} - \frac{1}{2}q\right) + D\left(x^2 - \frac{1}{2}qx^3 + \&c.\right),$$

the second member of which is in fact the series derived from the root $\alpha = 2$. This series is expressible by means of an exponential, viz. we have

$$x^2 - \frac{1}{2}qx^3 + \&c. = \frac{12}{q^3} \left\{ \left(\frac{1}{x} + \frac{1}{2}q\right) e^{-qx} - \left(\frac{1}{x} - \frac{1}{2}q\right) \right\},$$

and the complete integral is thus

$$y = A\left(\frac{1}{x} - \frac{1}{2}q\right) + B\left(\frac{1}{x} + \frac{1}{2}q\right) e^{-qx},$$

but this result is not immediately connected with the investigation. It may however be noticed that, writing $z = ye^{qx}$, the equation in z is

$$\frac{d^2z}{dx^2} - q\frac{dz}{dx} - \frac{2}{x^2}z = 0,$$

which only differs from the original equation in that it has $-q$ in place of q : there is consequently the particular solution $z = \frac{1}{x} + \frac{1}{2}q$, giving for y the particular solution $y = \left(\frac{1}{x} + \frac{1}{2}q\right) e^{-qx}$, and we have thence the complete solution as above.

Consider, secondly, the differential equation

$$\frac{d^2y}{dx^2} - \left(\frac{1}{4}q^2 + \frac{2}{x^2}\right)y = 0, \tag{II}$$

(derived from the equation (I) by writing therein ye^{-qx} in place of y); this equation is satisfied by the series

$$y = Ax^\alpha + Bx^{\alpha+2} + Cx^{\alpha+4} + \&c.$$

p. 461 [532]

Then $\alpha = -1$ or $\alpha = +2$, but here the series belonging to $\alpha = -1$ contains only odd powers, the other contains only even powers of x ; hence the two series do not coalesce as in the former case, and the first series is obtained without the indeterminate symbol $\frac{0}{0}$ in any of the coefficients. We have in fact

$$\begin{aligned}(\alpha + 1)(\alpha - 2)A &= 0, \\(\alpha + 3)\alpha B - \frac{1}{4}q^2 A &= 0, \\(\alpha + 5)(\alpha + 2)C - \frac{1}{4}q^2 B &= 0, \\&\&c.\end{aligned}$$

and there is not, in the series in question for the value of $\alpha = -1$ (or other series) any evanescent factor, either in the numerator or in the denominators.

But consider, thirdly, the differential equation

$$\frac{d^2 y}{dx^2} + (q - 2\theta)\frac{dy}{dx} + \left(\theta^2 - q\theta - \frac{2}{x^2}\right)y = 0, \quad (\text{III})$$

(derived from (I) by writing therein $ye^{-\theta x}$ in place of y). This is satisfied by the series

$$y = Ax^\alpha + Bx^{\alpha+1} + Cx^{\alpha+2} + Dx^{\alpha+3} + \&c.,$$

where $\alpha = -1$ or $\alpha = +2$ as before; in the series belonging to $\alpha = -1$, the coefficient D should become indeterminate. The relation between the coefficients is here a relation between three consecutive coefficients, viz. we have

$$\begin{aligned}(\alpha + 1)(\alpha - 2)A &= 0, \\(\alpha + 2)(\alpha - 1)B + (q - 2\theta)\alpha A &= 0, \\(\alpha + 3)\alpha C + (q - 2\theta)(\alpha + 1)B + (\theta^2 - q\theta)A &= 0, \\(\alpha + 4)(\alpha + 1)D + (q - 2\theta)(\alpha + 2)C + (\theta^2 - q\theta)B &= 0, \\(\alpha + 5)(\alpha + 2)E + (q - 2\theta)(\alpha + 3)D + (\theta^2 - q\theta)C &= 0, \\&\&c.\end{aligned}$$

It is to be shown that in the series for $\alpha = -1$, the expression $(q - 2\theta)(\alpha + 2)C + (\theta^2 - q\theta)B$ contains the evanescent factor $(\alpha + 1)$, and consequently that D is indeterminate; we have in fact

$$B = -\frac{(q - 2\theta)\alpha}{(\alpha + 2)(\alpha - 1)}A,$$

and thence

$$\begin{aligned}C &= \frac{1}{\alpha(\alpha + 3)} \left\{ \frac{(q - 2\theta)^2 \alpha (\alpha + 1)}{(\alpha + 2)(\alpha - 1)} - (\theta^2 - q\theta) \right\} A, \\&\quad (q - 2\theta)(\alpha + 2)C + (\theta^2 - q\theta)B\end{aligned}$$

p. 462 [532]

and then, after reduction, the remaining multiplier is

$$-\frac{\alpha+2}{\alpha(\alpha+3)} + \frac{\alpha}{(\alpha+2)(\alpha-1)} = \frac{2\alpha^3+6\alpha^2-4}{(\alpha-1)\alpha(\alpha+2)(\alpha+3)} = \frac{2(\alpha+1)(\alpha^2+2\alpha-2)}{(\alpha-1)\alpha(\alpha+2)(\alpha+3)},$$

so that the whole expression contains the factor $\alpha+1$. But observe that in the present case, if (as is allowable) we write $D=0$, the next coefficient E (depending not on D only, but on D and C) will not vanish; so that the solution obtained on the assumption $D=0$ will go on to infinity: and if instead of assuming $D=0$, we assume D = an arbitrary quantity D' , then E and the subsequent coefficients will contain terms depending on D' , and the complete form of the series belonging to $\alpha = -1$ will be

$$y = A \left(x^{-1} + \frac{B}{A} + \frac{C}{A}x + \frac{D}{A}x^2 + \frac{E}{A}x^3 + \&c. \right) + D' \left(x^2 + \frac{E'}{D'}x^3 + \&c. \right),$$

where the second member is in fact the series belonging to $\alpha = 2$. It is hardly necessary to remark that the solution thus obtained can be expressed by means of exponentials, viz. that the solution is

$$y = A \left(\frac{1}{x} - \frac{1}{2}q \right) e^{\theta x} + D' \frac{12}{q^3} \left\{ \left(\frac{1}{x} + \frac{1}{2}q \right) e^{-qx} - \left(\frac{1}{x} - \frac{1}{2}q \right) \right\} e^{\theta x}.$$

[533]

On the Binomial Theorem, Factorials, and Derivations.

[From the Oxford, Cambridge, and Dublin Messenger of Mathematics, vol. v. (1869), pp. 102-114.]

THE following was part of my course of lectures in the year 1867.

The proof commonly called “Euler’s” of the binomial theorem is as follows: the theorem is assumed to be true for positive integer indices; that is, it is assumed that for any positive integer m we have

$$(1+x)^m = 1 + mx + \frac{m(m-1)}{1 \cdot 2}x^2 + \&c.$$

This being so, since $(1+x)^m \cdot (1+x)^n = (1+x)^{m+n}$, the equation

$$\begin{aligned} & \left[1 + mx + \frac{m(m-1)}{1 \cdot 2}x^2 + \&c. \right] \cdot \left[1 + nx + \frac{n(n-1)}{1 \cdot 2}x^2 + \&c. \right] \\ &= 1 + (m+n)x + \frac{(m+n)(m+n-1)}{1 \cdot 2}x^2 + \&c. \end{aligned}$$

is true for any positive integer values whatever of m and n ; the equation is therefore true identically; and it is consequently true for all values whatever of m and n . But any function ϕ_m of m , satisfying the functional equation $\phi_m \phi_n = \phi_{m+n}$, is an m th power, $= C^m$ suppose; that is, we have

$$C^m = 1 + mx + \frac{m(m-1)}{1 \cdot 2}x^2 + \&c.$$

where C is a constant; viz. it is independent of m ; but the value of C will of course depend upon x ; and if, in order to determine it, we write $m = 1$, the equation gives $C = 1 + x$; that is, we have

$$(1+x)^m = 1 + mx + \frac{m(m-1)}{1 \cdot 2}x^2 + \&c.,$$

which is the binomial theorem in its general form.

p. 464 [533]

It is to be observed that there is not in the demonstration any employment of the so called “principle of equivalent forms,” we do not from the truth of an equation for positive integer values of m, n , infer the truth of it for any values whatever of m, n ; there is the intermediate step, that being true for the integers, it is true identically, and the identical truth of the equation depends on the circumstance that comparing on the two sides of the equation the coefficients of the successive powers x, x^2, x^3 , &c., these coefficients are in every case finite, rational, and integral functions of m, n .

For instance, comparing the coefficients of x^2 , we have

$$(m+n)(m+n-1) = m(m-1) + 2mn + n(n-1),$$

and any such equation, being true for all positive integer values of m, n , will be true identically; developing the two sides, the equation is in fact

$$m^2 + 2mn + n^2 - m - n = m^2 - m + 2mn + n^2 - n.$$

The reasoning is thus perfectly good; but I remark that it is quite as easy to prove the general equation of which the last mentioned equation is an example, as it is to prove the binomial theorem for positive integer indices; and consequently that we can without the aid of the binomial theorem for positive integer indices prove the fundamental equation

$$(1+x)^{m+n} = (1+x)^m(1+x)^n$$

To show this I introduce the factorial notation and it is readily found to be

$$m(m-1)\cdots(m-r+1) = [m]^r;$$

this being so, the equation obtained by comparing the coefficients of x^r is readily found to be

$$[m+n]^r = [m]^r + r[m]^{r-1}[n]^1 + \frac{r(r-1)}{1 \cdot 2}[m]^{r-2}[n]^2 + \&c.,$$

and I say that, this, the *factorial binomial theorem for a positive integer index r* , is proved as easily and in the same manner as the binomial theorem for a like value of the index; or say in equation

$$(m+n)^r = m^r + rm^{r-1} \cdot n + \frac{r(r-1)}{1 \cdot 2}m^{r-2} \cdot n^2 + \&c.$$

To show how this is, I form the values of $[m+n]^1, [m+n]^2, [m+n]^3$, &c. successively, by what may be termed the process of varied multiplication: we have

$$[m+n]^1 = m+n.$$

p. 465 [533]

to obtain $[m+n]^2$ we have to multiply this by $m+n-1$: in regard to the first term m , I write the multiplier under the form $(m-1)+n$, and in regard to the second term n , under the form $m+(n-1)$; the process then stands

$$\begin{aligned} [m+n]^1 &= m+n \\ \frac{m+n-1}{[m+n]^2} &= \frac{(m-1)+n \quad ; \quad m+(n-1)}{m(m-1)+mn+mn+n(n-1)} \\ &= [m]^2 + 2[m]^1[n]^1 + [n]^2. \end{aligned}$$

To form $[m+n]^3$ we have to multiply by $m+n-2$; in regard to the first term, this is written under the form $(m-2)+n$; in regard to the second term under the form $(m-1)+(n-1)$; and in regard to the third term under the form $m+(n-2)$; the product is thus obtained in the form

$$\begin{aligned} [m]^3 + [m]^2[n]^1 + 2[m]^2[n]^1 + 2[m]^1[n]^2 + [m]^1[n]^2 + [n]^3 \\ = [m]^3 + 3[m]^2[n]^1 + 3[m]^1[n]^2 + [n]^3; \end{aligned}$$

and so on; the law of the terms is obvious; and the numerical coefficients are in effect obtained as follows:

$$\begin{array}{ccccccc} & & & & 1, & & \\ & & & & 1, & & 1, \\ & & & & 1, & 1, & 1, \\ & & & 1, & 1, & & \\ & & & 1, & 2, & 1, & 1, \\ & & & & 1, & 2, & 1, \\ & & 1, & 3, & 3, & 1, & \\ & & & & & & \&c. \end{array}$$

viz. by precisely the same process as is used in finding the numerical coefficients in the powers $(m+n)^1$, $(m+n)^2$, $(m+n)^3$, &c.; and we thus see that for any integer value r of the index, we have a factorial binomial theorem, wherein the numerical coefficients are the same as in the binomial theorem for the same index.

p. 466 [533]

But the method of varied multiplication may be applied to the demonstration of a much more general theorem; viz. we may use it to develop a product such as

$$(h-a)(h-b)(h-c)(h-d)(h-e),$$

according to a series of products

$$(h-\alpha)(h-\beta)(h-\gamma)(h-\delta)(h-\epsilon),$$

$$(h-\alpha)(h-\beta)(h-\gamma)(h-\delta),$$

$$(h-\alpha)(h-\beta)(h-\gamma),$$

$$(h-\alpha)(h-\beta),$$

$$(h-\alpha),$$

$$1.$$

For this purpose, starting with

$$h-a = h-\alpha + \alpha-a,$$

we multiply by $h-b$, written first under the form $h-\beta + \beta-b$, and then under the form $h-\alpha + (\alpha-b)$; we have thus

$$\begin{aligned} (h-a)(h-b) &= (h-\alpha)(h-\beta) \\ &\quad + (h-\alpha)\{(\alpha-a) + (\beta-b)\} \\ &\quad + 1 \cdot (\alpha-a)(\alpha-b), \end{aligned}$$

and so on. It is easy to see that we may for instance write

$$\begin{aligned} (h-a)(h-b)(h-c)(h-d)(h-e) &= (h-\alpha)(h-\beta)(h-\gamma)(h-\delta)(h-\epsilon) \\ &\quad + (h-\alpha)(h-\beta)(h-\gamma)(h-\delta) \left(\begin{matrix} \alpha, \beta, \gamma, \delta, \epsilon \\ a, b, c, d, e \end{matrix} \right)_1 \\ &\quad + (h-\alpha)(h-\beta)(h-\gamma) \left(\begin{matrix} \alpha, \beta, \gamma, \delta \\ a, b, c, d, e \end{matrix} \right)_2 \\ &\quad + (h-\alpha)(h-\beta) \left(\begin{matrix} \alpha, \beta, \gamma \\ a, b, c, d, e \end{matrix} \right)_3 \\ &\quad + (h-\alpha) \left(\begin{matrix} \alpha, \beta \\ a, b, c, d, e \end{matrix} \right)_4 \\ &\quad + 1 \left(\begin{matrix} \alpha \\ a, b, c, d, e \end{matrix} \right)_5. \end{aligned}$$